Lecture L16

*Blind review — please score this lecture on row **L16** in the Excel form (1–10 per dimension).*

Backpropagation in Neural Networks

Learning Objectives:

1. Understand the mechanics of backpropagation and its role in training complex neural networks.
2. Apply the backpropagation algorithm to a simple neural network.
3. Recognize the limitations of backpropagation, including its computational expense and sensitivity to hyperparameters.

Why Backpropagation Matters:

Imagine a pipeline that carries information from the input to the output, critical for understanding how neural networks process information. Forward propagation is the process of computing the output of a neural network given an input x. Now, let's see how backpropagation helps optimize this process.

Formal Definition:

Forward propagation can be formally defined as the process of computing the output of a neural network given an input x.

Worked Example:

Consider a simple neural network with a single hidden layer. Computing the output of this network given an input x involves computing the output of the input layer, then the output of the hidden layer, and finally the output of the output layer. Try this yourself: can you think of a situation where forward propagation alone is not enough?

Misconception:

Some students might think that backpropagation is only used for training neural networks. However, backpropagation is also used for computing the gradient of the cost with respect to the model's parameters, which is essential for training. Can you think of a situation where backpropagation might not be the best algorithm for training a neural network?

Backpropagation: Unleashing the Power of Neural Networks

The back-propagation algorithm allows the information from the cost to flow backwards through the network, in order to compute the gradient. This algorithm is often simply called backprop (Rumelhart et al., 1986a). Computing an analytical expression for the gradient is straightforward, but numerically evaluating such an expression can be computationally expensive. Think of backpropagation as a pipeline that carries information from the cost to the model's parameters. This pipeline is critical for computing the gradient of the cost with respect to the model's parameters.

Formal Definition:

Backpropagation can be formally defined as the process of computing the gradient of the cost with respect to the model's parameters.

Worked Example:

Consider a simple neural network with a single hidden layer. Computing the gradient of the cost with respect to the model's parameters given a minibatch of examples involves computing the gradient of the cost with respect to the output of the output layer, then the gradient of the cost with respect to the

output of the hidden layer, and finally the gradient of the cost with respect to the output of the input layer. Try this yourself: can you compute the gradient of the cost with respect to the model's parameters for a simple neural network?

Example: Back-Propagation for MLP Training

As an example, we will walk through the back-propagation algorithm as it is used to train a multilayer perceptron (MLP). We will develop a very simple MLP with a single hidden layer. To train this model, we will use minibatch stochastic gradient descent. The back-propagation algorithm is used to compute the gradient of the cost on a single minibatch.

Limitations:

While backpropagation is a powerful algorithm, it is not without its limitations. For example, backpropagation can be computationally expensive, particularly for large neural networks. Additionally, backpropagation can be sensitive to the choice of hyperparameters, such as the learning rate and the number of hidden layers.

Conclusion:

In this lecture, we have explored the concept of backpropagation in neural networks. We have seen how backpropagation allows us to train complex neural networks using gradient-based optimization techniques. We have also seen some of the limitations of backpropagation, such as its computational expense and its sensitivity to hyperparameters. By understanding the mechanics of backpropagation, we can better appreciate the power and limitations of deep learning algorithms.